

Remote Login

Remote Login

1. Serial Port Login
2. Network Status Confirmation
3. NoMachine Login

Connecting to the Development Board from Windows

4. SSH Login
 - Terminal Software
 - Computer Command Line

This section aims to introduce users who need to remotely access the development board from a personal computer (PC) to methods of remote login via serial port and network (SSH).

Before logging in remotely via the network, the development board needs to be connected to the network via wired Ethernet or wireless Wi-Fi, and the development board's IP address must be configured. For IP address information under the two connection methods, refer to the description below:

- Wired Ethernet: Board uses static IP mode by default, with IP address `192.168.127.10`, mask `255.255.255.0`, gateway `192.168.127.1`
- Wireless Wi-Fi: The development board's IP address is generally assigned by the router. You can check the IP address of the wlan0 network using the `ifconfig` command in the device command line.

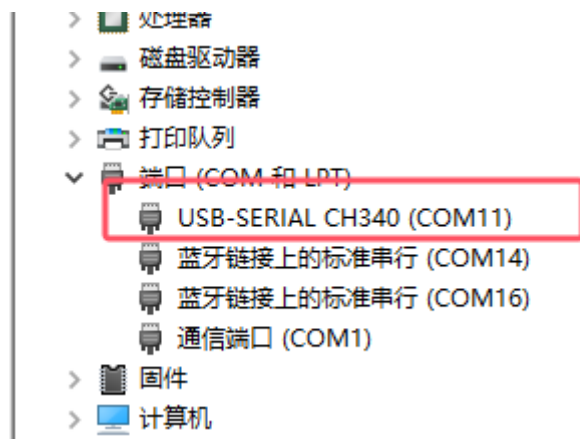
1. Serial Port Login

Before logging in via the serial port, you need to ensure that the development board's serial port cable is correctly connected to the computer.



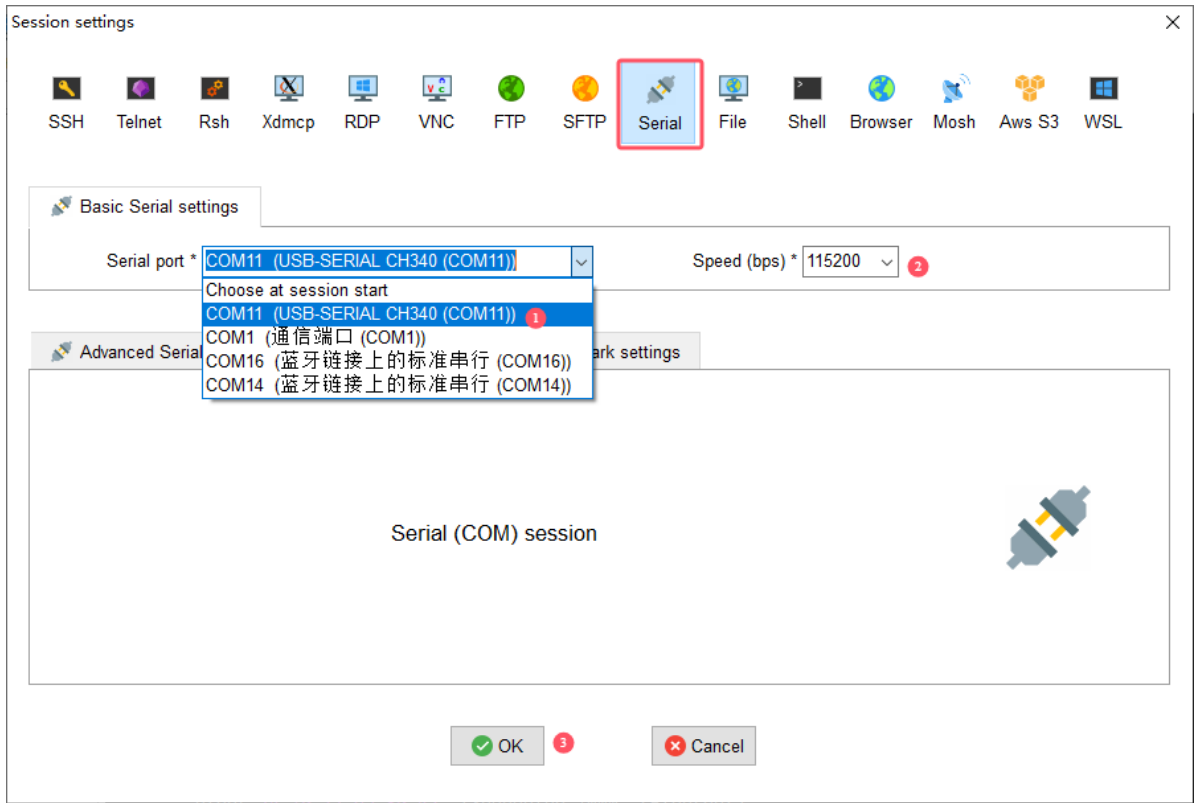
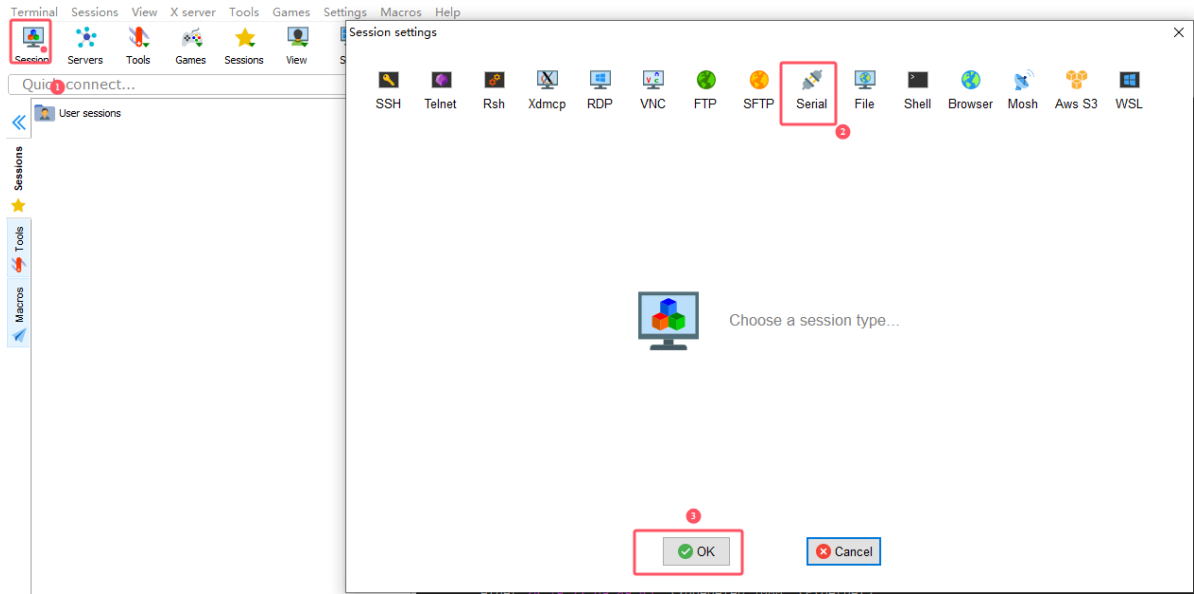
Serial port login requires a PC terminal tool. Commonly used tools include `Putty`, `MobaXterm`, etc. Users can choose according to their habits. The port configuration process for different tools is basically similar. The following uses `MobaXterm` as an example to introduce the process of creating a new serial port connection:

- When the USB-to-serial adapter board is first plugged into the computer, the CH340 serial port driver needs to be installed. After driver installation is complete, the serial port will be correctly recognized in Device Manager, as shown below:



- Open the `mobaXterm` tool, click `session`, then select `Serial`
- Configure the port number, e.g., `COM11`. The actual serial port number used depends on the port recognized by the PC.
- Set the serial port configuration parameters as follows:

Configuration Item	Parameter Value
Baud rate	921600
Data bits	8
Parity	None
Stop bits	1
Flow Control	None



Click **OK**, log in to the device by entering username: `root`, password: `root`

```

ubuntu login:
ubuntu login:
ubuntu login: root
Password:
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.1.83 aarch64)

* Documentation: https://help.ubuntu.com
* Management:   https://landscape.canonical.com
* Support:      https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

36 updates can be applied immediately.
15 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

62 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

root@ubuntu:~# █

```

At this point, you can use the `ifconfig -a` command to query the development board's IP address, where eth0/eth1 and wlan0 represent wired and wireless networks respectively:

```

eth0: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
    ether 7a:5e:ca:06:4b:a1 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
    device interrupt 136

eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.127.10 netmask 255.255.255.0 broadcast 192.168.127.255
    inet6 fe80::58de:11ff:fe64:e19c prefixlen 64 scopeid 0x20<link>
    ether 5a:de:11:64:e1:9c txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 3 bytes 425 (425.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
    device interrupt 192

eth2: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
    ether 02:8f:6f:81:99:10 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
    device interrupt 201

eth3: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
    ether ee:71:51:40:ac:ad txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

```

```
device interrupt 218
```

```
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 224 bytes 21518 (21.5 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 224 bytes 21518 (21.5 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

wlan0: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
    ether f0:68:e3:22:7e:90 txqueuelen 1000 (Ethernet)
    RX packets 8280 bytes 654536 (654.5 KB)
    RX errors 0 dropped 5898 overruns 0 frame 0
    TX packets 1138 bytes 139677 (139.6 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

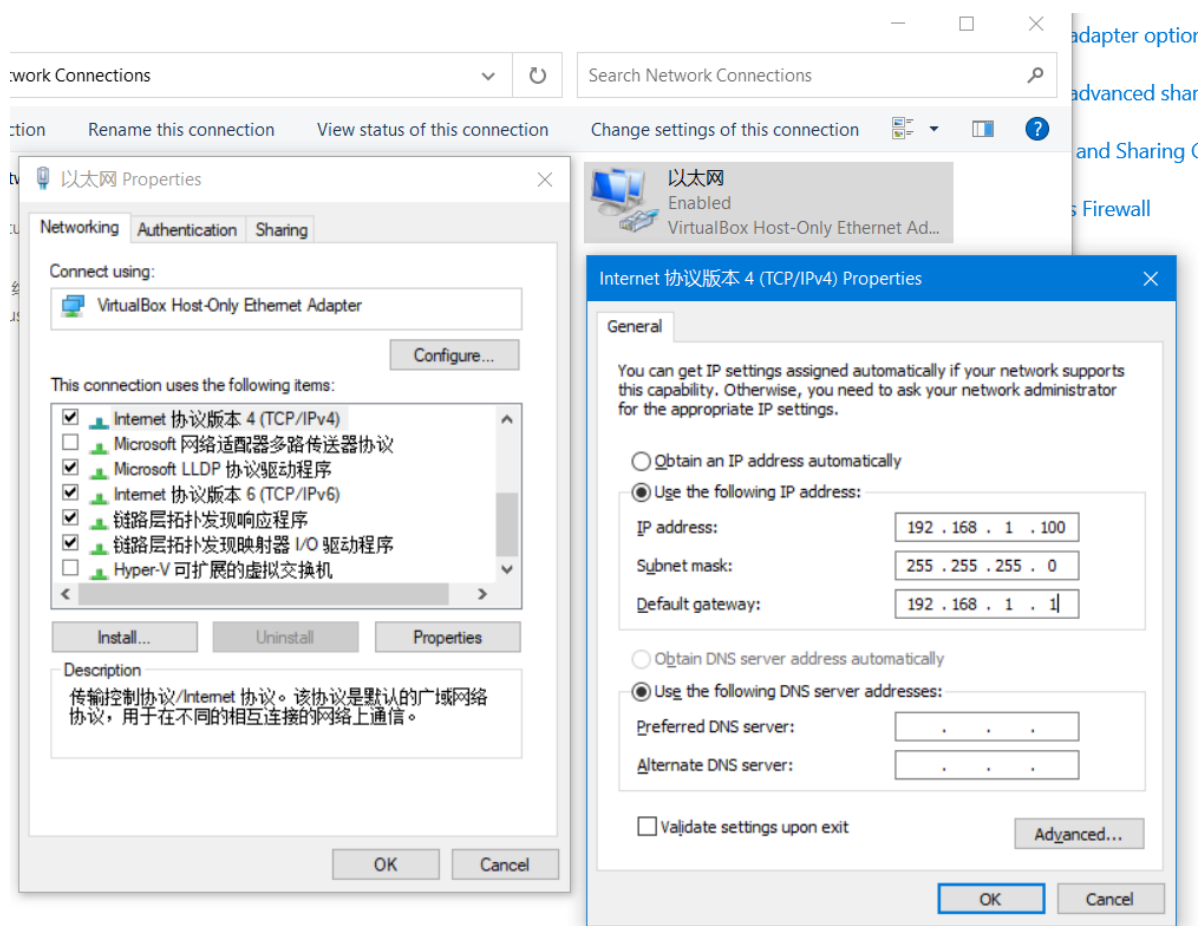
2. Network Status Confirmation

Before using remote login, you need to ensure that the computer and development board have normal network communication. If you cannot ping them, follow these steps to confirm:

- Confirm the IP address configuration of the development board and computer. Generally, the first three segments need to be the same, for example: Development board: 192.168.127.10 Computer: 192.168.127.100
- Confirm that the subnet mask and gateway configuration of the development board and computer are consistent.
- Confirm whether the computer's network firewall is turned off.

The outer wired Ethernet port (eth1) of the development board uses static IP mode by default, with IP address 192.168.127.10. For direct network connection between the development board and computer, simply configure the computer with a static IP, ensuring it is on the same network segment as the development board. Taking the WIN10 system as an example, the method to modify the computer's static IP is as follows:

- Find the corresponding Ethernet device in Network Connections and double-click to open it.
- Find the Internet Protocol Version 4 option and double-click to open it.
- Fill in the corresponding network parameters in the red box area in the figure below and click OK.



If you need to configure the development board's wired network to DHCP mode for dynamic acquisition, refer to the 【Network Configuration】 section.

3.NoMachine Login

The Yahboom image comes with NoMachine pre-installed. Users can achieve remote control by installing NoMachine on Windows.

This chapter is intended for users of the S600 running the Ubuntu Desktop system version, and describes how to use `NoMachine` for remote desktop login.

S600 NoMachine Configuration

`NoMachine` currently does not support `apt` download. You need to obtain the `deb` package from the official website.

Official NoMachine download URL: [NoMachine Download](#)

After entering the official website, find the `ARM64` version of the installation package suitable for `RDK S600` and click `Download`.

ARM 64-bit



NoMachine for ARM RPM (aarch64)

Version: 9.0.188_11
Package size: 65.86 MB
Package type: RPM
MD5 signature: 6f8a7135c03bb2d3b3e563a33ea6bad5
For: Fedora 26/27/28/29/30/31/32/33/34, RHEL 7/8, OpenSUSE Tumbleweed

Download



NoMachine for ARM DEB (arm64)

Version: 9.0.188_11
Package size: 65.12 MB
Package type: DEB
MD5 signature: a895cee274347e8df83de1d66fdd2007
For: Ubuntu 14.04/16.04/18.04/20.04, Debian 8/9/10

Download



NoMachine for ARM TAR.GZ (aarch64)

Version: 9.0.188_11
Package size: 65.24 MB
Package type: TAR.GZ
MD5 signature: 18b24f6c6df2938eb9af8dbeb2add102
For: Ubuntu 14.04/16.04/18.04/20.04, Debian 8/9/10, Fedora 26/27/28/29/30/31/32/33/34, RHEL 7/8, OpenSUSE Tumbleweed

Download

Installation

```
sudo apt update; sudo apt upgrade -y // Ensure the current software is up to date
dpkg -i nomachine-*_arm64.deb
```

Configuration and Startup

1. Configure the server to allow remote connections:

```
sudo systemctl start nxserver
```

2. Set `NoMachine` to start on boot:

```
sudo systemctl enable nxserver
```

3. Set `EGL capture` to `yes`. This is a screen capture feature provided by `NoMachine`, mainly used to improve the remote desktop experience in specific display server environments:

```
sudo /etc/NX/nxserver --eglcapture yes
```

This command takes effect after a reboot. You can double-check with the following command; `EGL capture has been enabled` indicates that the feature has been written to the configuration file.

```
if [ -f "/usr/lib/systemd/user/org.gnome.Shell@wayland.service" ] && grep -q
"nxpreload.sh" "/usr/lib/systemd/user/org.gnome.Shell@wayland.service" && [
-f "/usr/share/applications/org.gnome.Shell.desktop" ] && grep -q
"nxpreload.sh" "/usr/share/applications/org.gnome.Shell.desktop" && [ -f
"/usr/NX/etc/node.cfg" ] && grep -q "EnableEGLCapture 1"
"/usr/NX/etc/node.cfg"; then echo "EGL Capture has been enabled"; else echo
"Not enabled"; fi
```

4. Restart the `NoMachine` service:

```
sudo systemctl restart nxserver
```

Restart the S600

Due to `NXServer` configuration issues, connecting directly after performing the above operations will result in a black screen. A reboot is required before use.

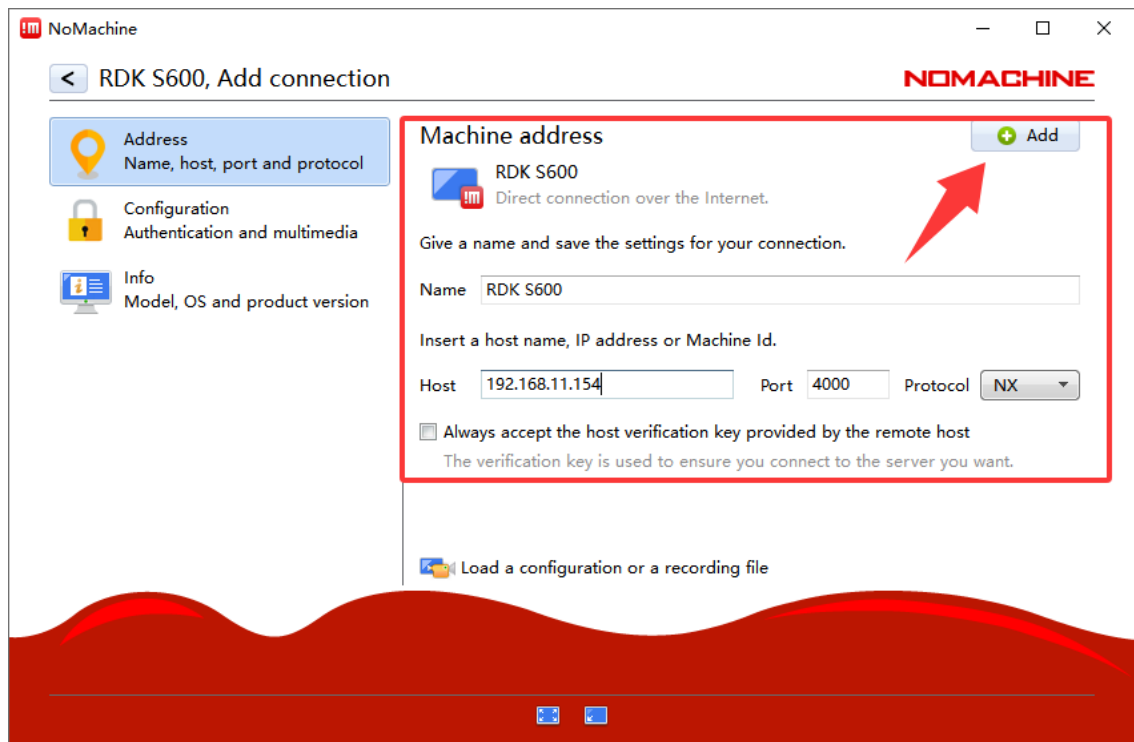
Connecting to the Development Board from Windows

Before connecting to the development board, ensure that NoMachine is installed on Windows. You can install it using the installer provided in the attachments folder, or download it from the [official website](#).

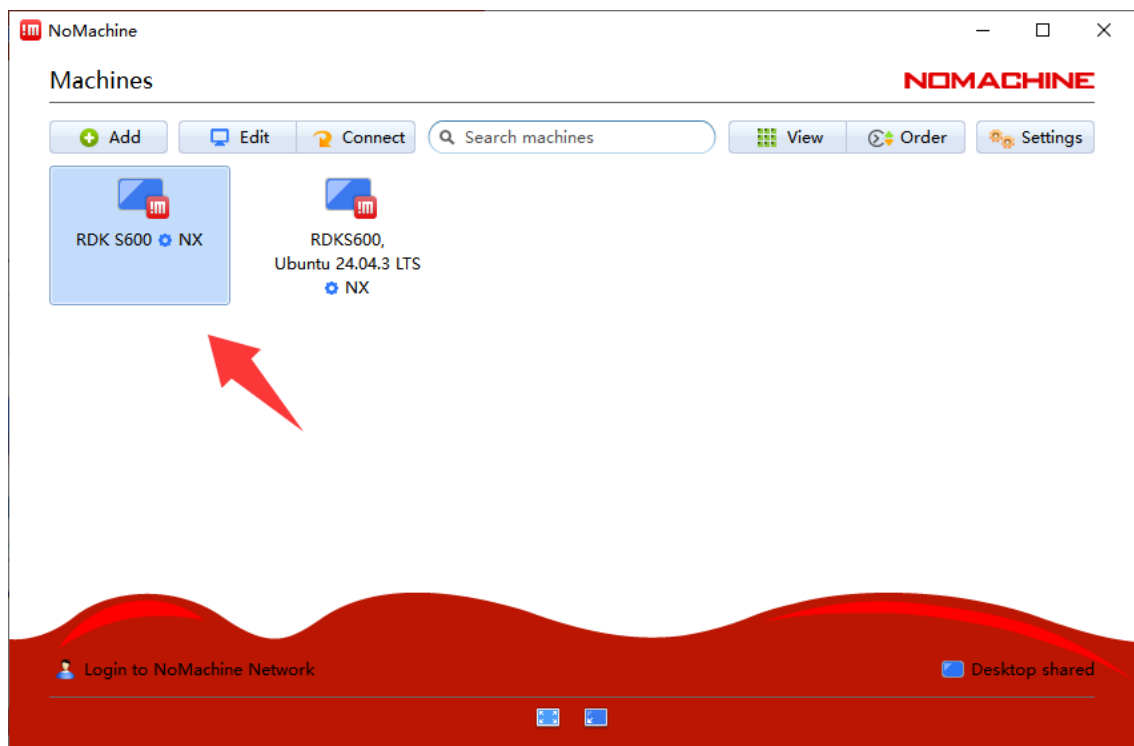
1. Open the `NoMachine` client and click `Add` to add a host configuration.



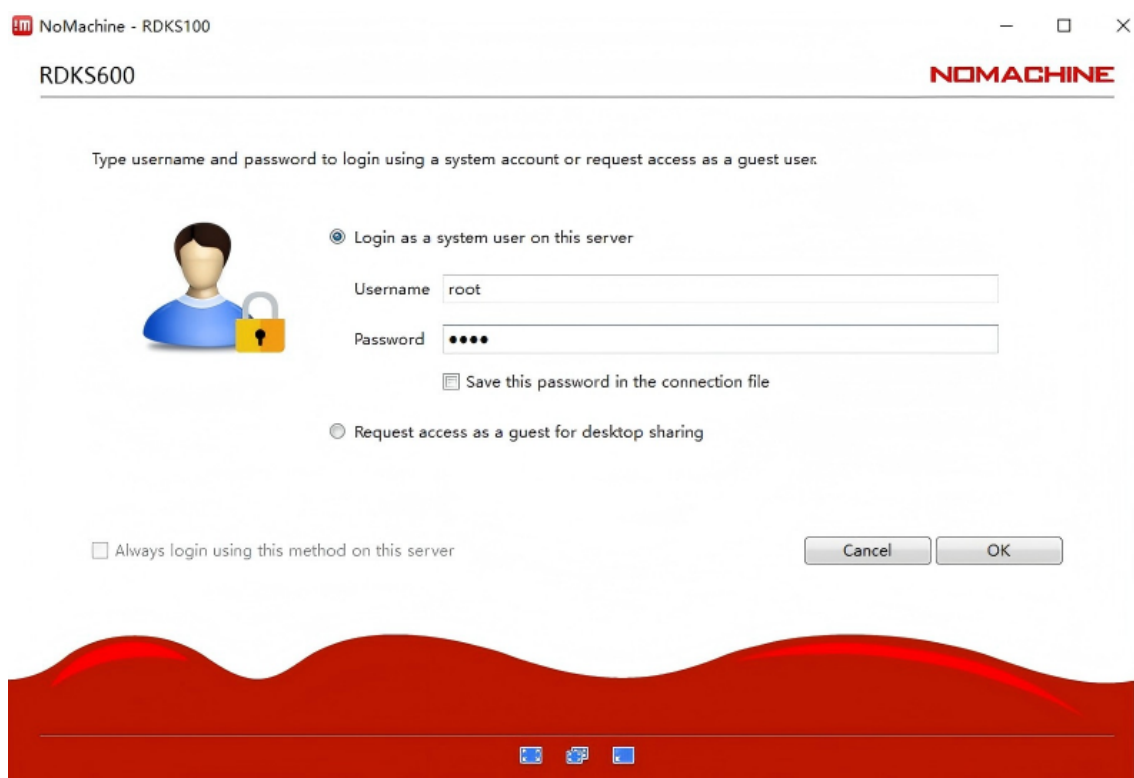
2. In the pop-up interface, fill in the host information for `RDK S600`, then click `Add`.

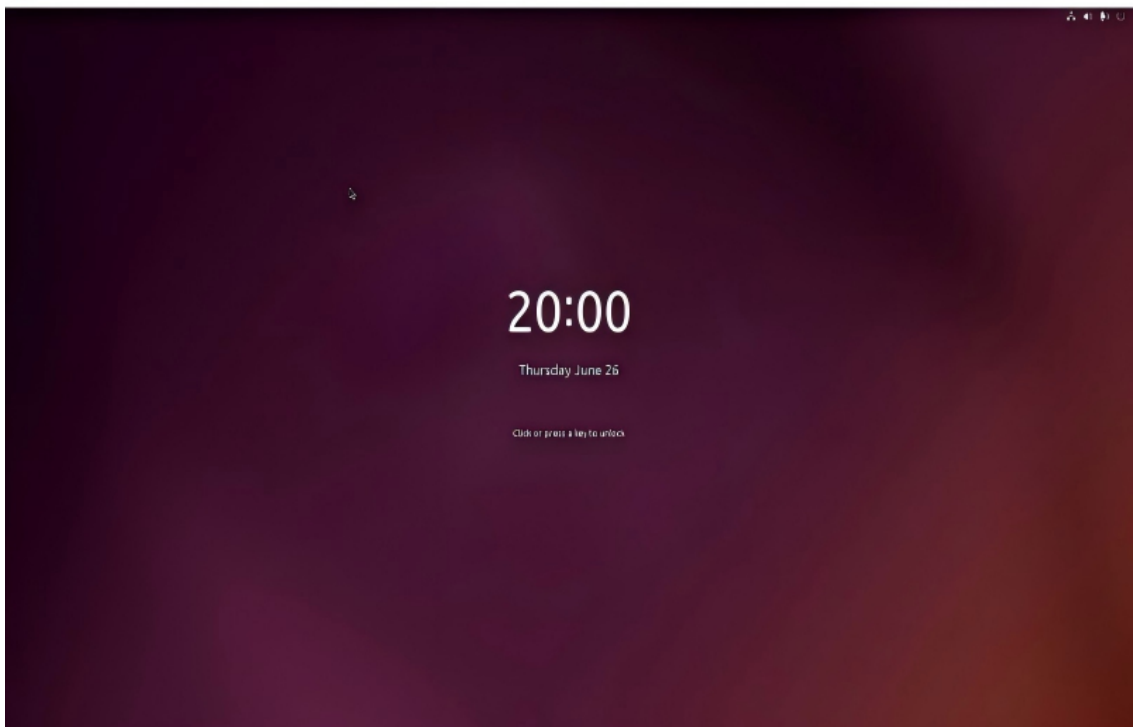


3. Return to the main interface and double-click the host you just created.



4. A login interface will appear. Enter the username and password, then click OK to complete the remote login.



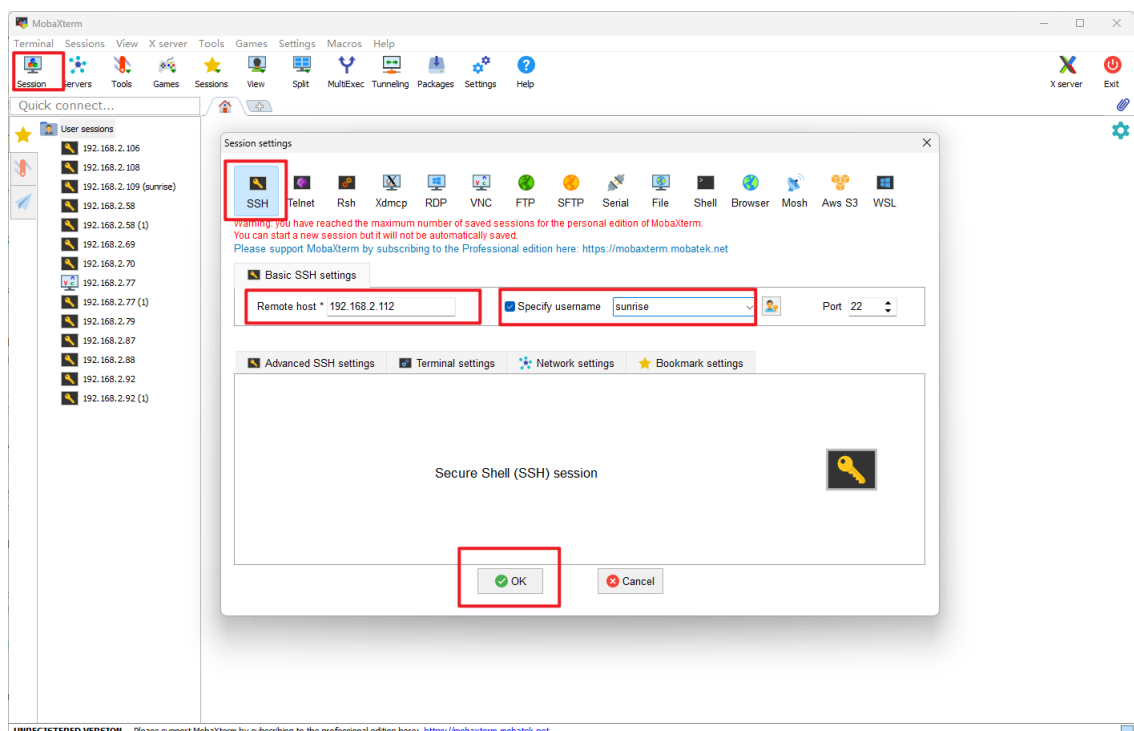


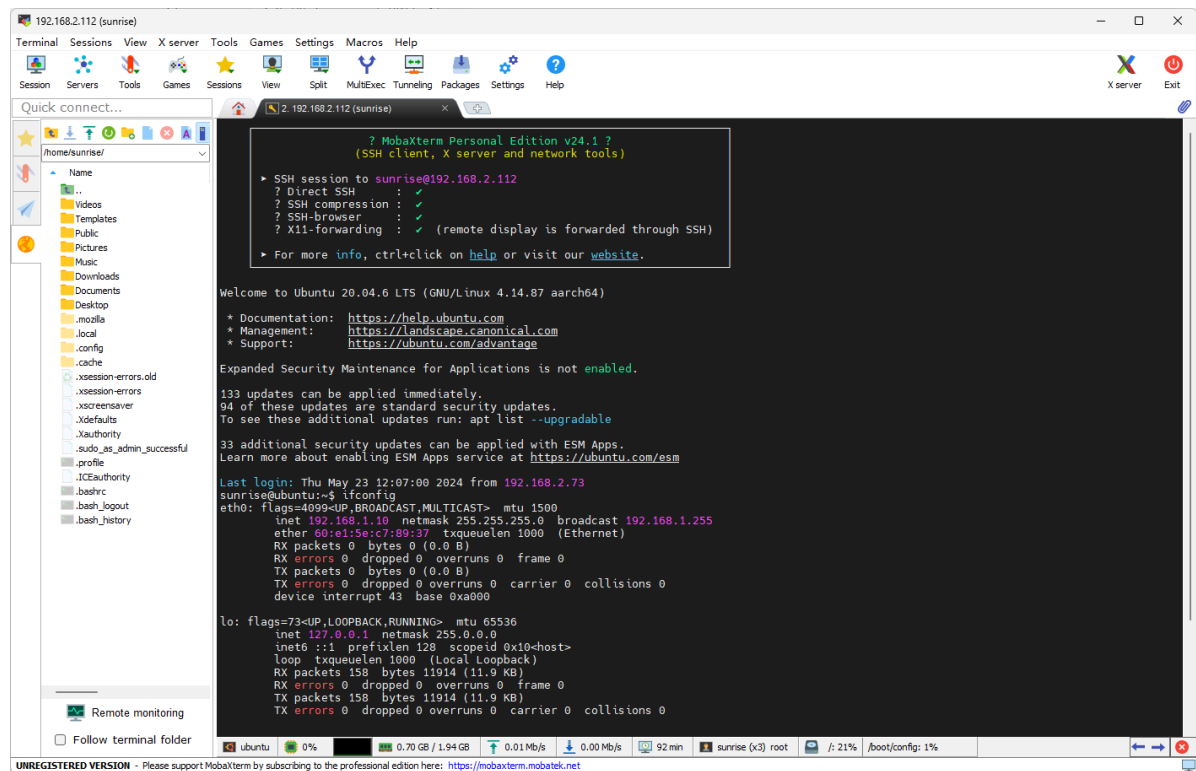
4. SSH Login

Terminal Software

Commonly used terminal tools include `Putty`, `MobaXterm`, etc. Users can choose according to their habits. The port configuration process for different tools is basically similar. The following uses `MobaXterm` as an example to introduce the process of creating a new SSH connection:

- Open the `MobaXterm` tool, click `Session`, then select `SSH`
- Enter the development board's IP address, e.g., `192.168.127.10`
- Select `specify username` and enter `sunrise`
- After clicking OK, enter the username (sunrise) and password (sunrise) to complete the login





Computer Command Line

Users can also log in via SSH using the command line. The steps are as follows:

1. Open a terminal window and enter the SSH login command, e.g., `ssh sunrise@192.168.127.10`
2. When a connection confirmation prompt appears, enter YES
3. Enter the password (sunrise) to complete the login